## 1. Project Objective

The objective of this project is to build a shared enterprise AI Knowledge Hub. The platform will transform company documents and information into searchable knowledge bases that AI systems can use effectively.

When a user asks a question, the AI should first retrieve the most relevant enterprise knowledge and then generate a grounded answer based on those sources. The platform should also provide quality validation, usage monitoring, and token/cost transparency so that the solution can be evaluated and improved over time.

## 2. Minimum Required Scope for This Semester

Based on the sponsor discussion, the yellow-highlighted capabilities are the minimum required scope. Enterprise-scale features outside this area may be documented as future extensions, but they should not delay the core MVP.

| Workstream | What We Will Build | Plain-Language Meaning | Expected Demonstrable Outcome |
|---|---|---|---|
| . Knowledge Ingestion & Knowledge Base Creation | Analyze uploaded documents, identify content/type, create metadata, chunk content, and build indexes. | Turn ordinary enterprise files into knowledge that AI can search and use. | Upload a document and automatically create a searchable knowledge base. |
| . Retrieval & Search | Build hybrid retrieval using keyword/full-text search, semantic/vector search, and re-ranking. | Find both exact terms and conceptually similar information, then select the most relevant evidence. | A user question retrieves relevant knowledge of chunks that can be inspected and used for answering. |
| . Quality Validation & Testing | Create certified questions, Judge Agent, quality/freshness metrics, source traceability, and feedback mechanisms. | Do not accept an answer only because it sounds reasonable; measure whether it is supported, complete, relevant, and current. | Quality results show whether retrieved evidence and generated answers meet defined standards. |
| . Monitoring, Administration & Cost Transparency | Track knowledge bases, usage, token consumption, estimated cost, and basic operational metrics. | Understand how the system is being used, how much it costs, and whether the quality justifies the cost. | A monitoring view shows query activity, token usage, estimated cost, KB status, and key quality indicators. |

## 3. Detailed Workstreams

### Workstream 1: Knowledge Ingestion & Knowledge Base Creation

- Support document ingestion such as PDF, TXT, and MD files.
- Analyze document content and source type.
- Generate relevant metadata such as topic, source, and document type.
- Apply chunking so long documents are divided into retrieval-friendly sections.
- Create indexes that allow later queries to find information efficiently.

Success criterion: The team can demonstrate the complete flow from document upload to searchable enterprise knowledge.

### Workstream 2: Retrieval & Search

- Use keyword/full text to search for exact names, codes, acronyms, and technical terms.
- Use semantic/vector search to find conceptually similar information even when wording differs.
- Combine both approaches into hybrid retrieval.
- Re-rank candidate results, so the most relevant evidence is prioritized.
- Send only the most relevant context to the LLM whenever possible to balance quality and cost.

Success criterion: a user's question returns useful evidence from the knowledge base before an answer is generated.

### Workstream 3: Quality Validation & Testing

- Create a certified question set with known expected answers and/or expected source documents.
- Use a Judge Agent to evaluate whether an answer is supported by retrieved evidence.
- Measure retrieval of relevance, missed information, answer relevance, and faithfulness to sources.
- Track provenance and content freshness to reduce use of stale or unsupported information.
- Capture user feedback as an input to future retrieval and knowledge-based improvements.

Success criterion: the platform can explain not only what the answer is, but also why the team should trust it or where quality gaps remain.

### Workstream 4: Monitoring, Administration & Cost Transparency

- Maintain knowledge-based inventory and basic administration capability.
- Allow knowledge bases to be enabled or disabled for grounding.
- Track token usage, estimated cost, query activity, and knowledge-based usage.
- Evaluate cost together with answer quality rather than optimizing cost alone.
- Preserve an upgrade path for future adaptive cost-quality routing based on historical performance.

Success criterion: the team can answer how the system is being used, what it costs, and whether the output quality is sufficient.

## 4. End-of-Semester MVP Demo Flow

1. Upload enterprise documents.
2. Automatically analyze, chunk, enrich, and index the content.
3. Create a searchable knowledge base.
4. Enter a user question.
5. Use hybrid retrieval to identify the most relevant evidence.
6. Generate a grounded answer with a source of visibility.
7. Evaluate retrieval and answer quality.
8. Display token usage, estimated cost, usage, and quality information.

## 5. Out of Minimum Scope / Stretch Goals

Advanced graph-based relationship retrieval (for example, Supplier → Part → Product → Customer).

Full spreadsheet/SQL dynamic query capability.

Production-grade enterprise security, Azure deployment, and full RBAC implementation.

Fully automated historical learning and cost optimization; the semester MVP may focus on logging, analysis, and recommendations first.

Broad enterprise API/MCP integrations; if time allows, demonstrate one focused integration or action.

## 6. Initial Team Roles

| eam Member | rimary Role | rimary Responsibilities |
|---|---|---|
| enson | hared Project Management | cope management, milestones, neeting structure, sponsor alignment, ocumentation, progress tracking |

| | | |
|---|---|---|
| Nityanand | Shared Project Management | Scope management, milestones, meeting structure, sponsor alignment, documentation, progress tracking |
| Ethan | Full-Stack Developer | Back-end/front-end development, technical implementation, core integration |
| Erynn | Sponsor Point-of-Contact + Full-Stack Developer | Primary communication with Samir, requirement clarification, development and integration |

## 7. Semester Milestones & Deliverables

| Date | Milestone | Main Work | Deliverable | Scope Note |
|---|---|---|---|---|
| Sep 15–27 | M1 Scope & Architecture Alignment | Confirm minimum features, data sources, system flow, technical direction, and team responsibilities. | Scope document + initial architecture diagram | Minimum-scope design |
| Sep 28–Oct 4 | M2 Phase 1 Package | Finalize business context, requirements, data needs, scope, and proposed architecture. | Draft Phase 1 deck | Minimum-scope definition |
| Oct 6 | M3 Phase 1 Submission | Submit the course Phase 1 deliverable. | Project Phase 1 PPT | Course deadline |
| Oct 7–18 | M4 Knowledge Pipeline MVP | Implement document ingestion, analysis, chunking, indexing, and basic KB creation. | Searchable knowledge base | Core build |
| Oct 19–31 | M5 Retrieval Layer | Implement hybrid retrieval, re-ranking, and the basic question-to-evidence workflow. | Question → retrieval → relevant evidence | Core build |
| Nov 1–14 | M6 Quality Validation | Create certified questions, Judge Agent, Provenance, and Quality Metrics. | Validation result's / QA view | Core build |
| Nov 15–22 | M7 Monitoring & Cost | Implement token/cost tracking, usage monitoring, KB inventory/status. | Monitoring and cost view | Core build |
| Nov 23–30 | M8 End-to-End Integration | Connect ingestion, retrieval, answer generation, validation, and monitoring. | End-to-end Beta Prototype | Minimum-scope integration |
| Dec 1–4 | M9 Sponsor Demo | Demonstrate the minimum required scope to the sponsor and collect final feedback. | Sponsor-ready demo | Minimum-scope acceptance |
| Dec 5–7 | M10 Finalization | Apply fixes and finalize slides, code links, architecture, and lessons learned. | Final package | Finalization |

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|---|---|---|---|---|
| )ec 8 | M11 Phase 2 + Final )emo | Submit final project naterials and deliver the n-class demo. | Phase 2 PPT + Final )emo | Course deadline |

## 8. Project Management Principles

- Complete the yellow minimum required scope before investing significant time in stretch features.
- Each milestone should produce something reviewable or demonstrable, not only “research completed.”
- Use a consistent sponsor meeting structure: Progress, Demo, Decisions Needed, Risks, and Next Milestone.
- Choose technology based on what can be completed, tested, and demonstrated within the semester—not on enterprise complexity alone.
- Any cost optimization must preserve the minimum required answer-quality threshold.